

Aspirin use, lipoprotein(a), and calcific aortic valve disease: the Multi-ethnic Study of Atherosclerosis

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Abstract

Background and Aims

Lipoprotein(a) [Lp(a)] and LDL cholesterol (LDL-C) are causally linked to aortic valve calcium (AVC) and aortic stenosis (AS). Lipoprotein(a) has anti-fibrinolytic properties; therefore, aspirin may reduce cardiovascular disease risk among individuals with high Lp(a). This analysis sought to determine the association of aspirin with incident AVC and AS across Lp(a) and LDL-C levels.

Methods

This observational study included up to 6598 participants in the Multi-Ethnic Study of Atherosclerosis. Aortic valve calcium was measured on non-contrast cardiac computed tomography. Multivariable Cox hazards regression assessed the association of self-reported regular aspirin use (≥ 3 days/week) with incident AVC and severe AS, stratified by Lp(a) and LDL-C. Aortic valve calcium and Lp(a) values were not reported to participants.

Results

Mean age was 62 years, 53% were women, 23% reported regular aspirin use, 8% developed AVC (median 8.9 years), and 1% developed severe AS (median 16.7 years). Among individuals with elevated Lp(a), regular aspirin use was associated with a lower risk of incident AVC (Lp(a) ≥ 75 mg/dL: hazard ratio (HR) .42, 95% confidence interval (CI) .19–.93; Lp(a) ≥ 100 mg/dL: HR .17, 95% CI .04–.67) and severe AS (Lp(a) ≥ 50 mg/dL: HR .13, 95% CI .04–.47; Lp(a) ≥ 75 mg/dL: HR .02, 95% CI .001–.29). For participants with elevated LDL-C, there was no association of regular aspirin use with incident AVC (LDL-C ≥ 130 mg/dL: HR 1.02, 95% CI .66–1.58; LDL-C ≥ 160 mg/dL: HR 1.51, 95% CI .53–4.28) or severe AS (LDL-C ≥ 100 mg/dL: HR .70, 95% CI .39–1.26; LDL-C ≥ 130 mg/dL: HR .46, 95% CI .14–1.47).

Conclusions

In this exploratory analysis of prospective observational cohort data, regular aspirin use was associated with a lower risk of AVC and severe AS in persons with high Lp(a), but not high LDL-C. Confirmatory studies are required to determine the role of aspirin in the prevention of AVC and AS for persons with high Lp(a).

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Structured Graphical Abstract

Key Question

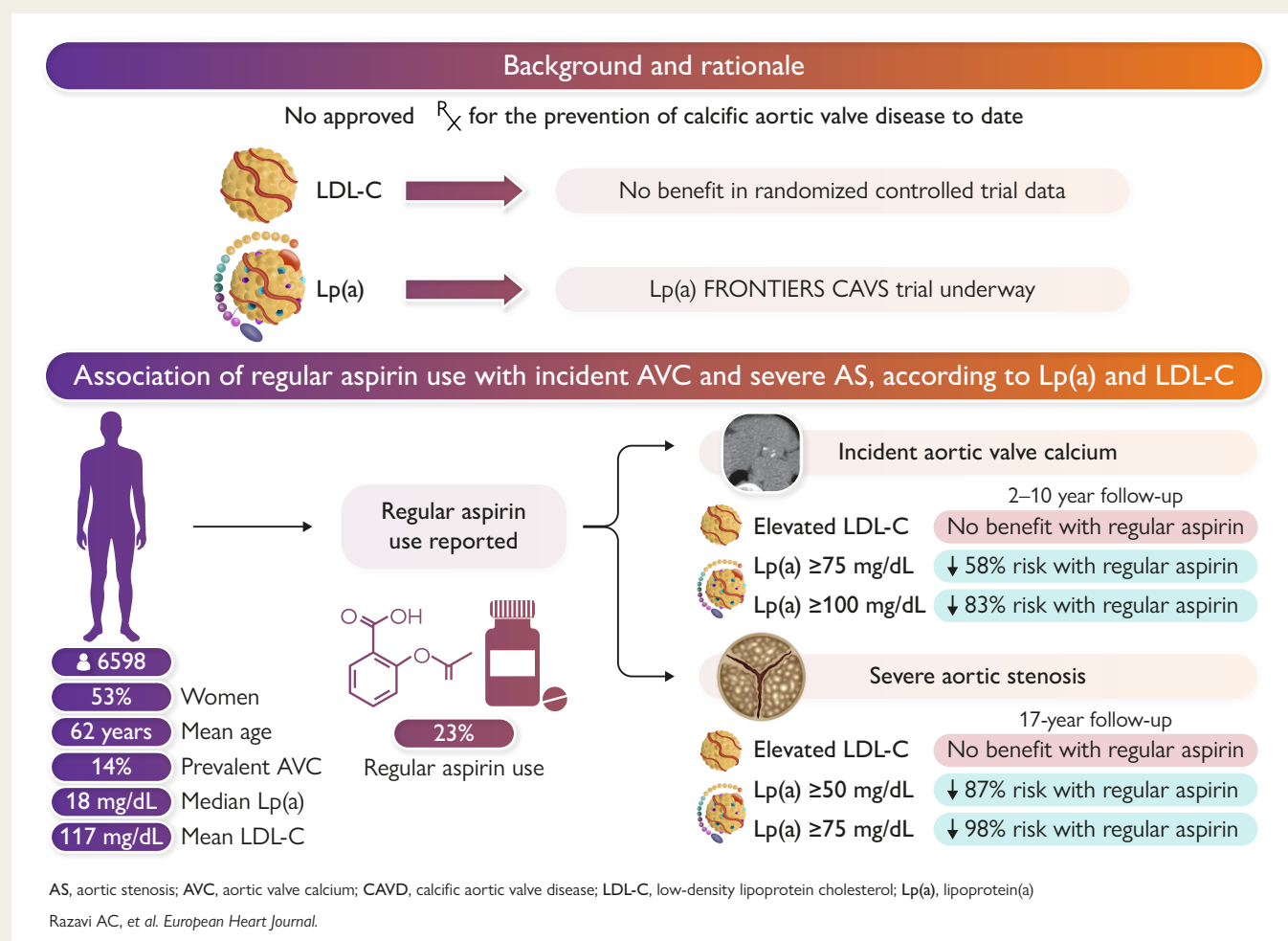
Can aspirin reduce the risk of incident aortic valve calcium (AVC) and severe aortic stenosis (AS)?

Key Finding

In this observational study including participants in the Multi-Ethnic Study of Atherosclerosis (MESA), regular aspirin use was associated with up to 83% lower risk for incident AVC and 98% lower risk for incident AS among individuals with elevated lipoprotein(a), but not elevated LDL-cholesterol.

Take Home Message

There are currently no approved therapies to slow the progression of calcific aortic valve disease. These innovative results require confirmation in randomized controlled trials.



Aspirin use, lipoprotein(a), low-density lipoprotein cholesterol, and calcific aortic valve disease.

*In this observational analysis, regular aspirin use was associated with a lower risk of AVC and severe AS in persons with high Lp(a), but not high LDL-C.

Keywords

Aspirin • Lipoprotein(a) • Aortic valve disease • Aortic stenosis • Aortic valve calcium • LDL cholesterol • Computed tomography

Introduction

Calcific aortic valve disease (CAVD) is the most common valvular heart disease in developed countries, especially among older adults.¹ Given an ageing population, the prevalence of CAVD

in the USA has increased more than two-fold over the last three decades, now affecting ~3.3 million individuals.¹ Individuals with aortic valve calcium (AVC), measured on non-contrast cardiac computed tomography (CT), experience a significantly higher long-term risk for severe aortic stenosis (AS), which

exponentially increases with higher AVC burden.² However, CAVD is one of the only cardiovascular diseases (CVDs) without a preventive therapy.

LDL cholesterol (LDL-C) and lipoprotein(a) [Lp(a)] are causal risk factors for the development of AVC and AS.^{3,4} Among all CVD outcomes, Lp(a) is most strongly associated with CAVD as there is a 1.5–2-fold higher risk for the presence of AVC for each 50 mg/dL increase in Lp(a).⁵ Similarly, over nearly one decade of follow-up, individuals with elevated Lp(a) have a 2.5-fold higher risk for incident AVC.⁶ In addition to its atherogenic properties, Lp(a) is also anti-fibrinolytic and proinflammatory.^{7,8} *In vitro*, Lp(a) and recombinant apolipoprotein(a) [apo(a)] more strongly induce platelet aggregation compared with LDL-C.⁹ Platelets are particularly important in the pathobiology of CAVD, because they accelerate fibrocalcific deposition via platelet-induced activation of osteogenic pathways.¹⁰

Recently, it has been shown that aspirin may be associated with a lower risk of CVD events among individuals with elevated Lp(a),^{11–16} which is thought to be due to the pro-thrombotic activity of Lp(a). Given such findings and the role of platelets in CAVD, we hypothesized that aspirin may reduce the risk of developing AVC and AS. However, there are no studies to assess the association of aspirin use with incident AVC and AS across the spectrum of Lp(a) values. The Multi-Ethnic Study of Atherosclerosis (MESA) is an observational cohort ideally suited to examine this hypothesis, as at the time MESA Visit 1 was performed, participants and their prescribing clinicians were not aware of AVC nor Lp(a) values and aspirin was prescribed according to individual participant and clinician judgement. Accordingly, we sought to examine whether there is any difference in risk for incident AVC and AS according to Lp(a) and LDL-C values among MESA participants who were and were not prescribed aspirin therapy.

Methods

Study population

Multi-Ethnic Study of Atherosclerosis is a community-based observational prospective cohort study, and specific details on the design and rationale have previously been reported.¹⁷ In short, MESA enrolled 6814 adults of White, Black, Hispanic, and Chinese race/ethnicity who were aged 45–84 years old and free of known clinical CVD. There were 6598 participants who had available measures of aspirin intake, Lp(a), and LDL-C and were followed for incident severe AS through 17-year follow-up. Among this cohort, there were 5066 participants with AVC = 0 at baseline who underwent repeat AVC scoring on cardiac CT 2–10 years later for the incident AVC analysis (see [Supplementary data online, Figure S1](#)).

All study participants provided written informed consent at each study visit, and study protocols were approved at each MESA participating institution's local Institutional Review Board (IRB) and sponsored by the National Heart, Lung, and Blood Institute (<http://www.mesa-nhlbi.org>).

Measurement of aortic valve calcium on cardiac computed tomography

Multi-Ethnic Study of Atherosclerosis participants underwent non-contrast cardiac CT at baseline (2000–02), which is a sensitive and specific modality for quantifying AVC.^{18,19} Cardiac CT scans were read at the Core MESA Laboratory without knowledge of any participant risk factor or medical history. Two scans were performed,

and the scores were averaged.²⁰ Calcified lesions on or within aortic valve leaflets of ≥ 3 continuous pixels that had a peak attenuation of ≥ 130 Hounsfield units valve contributed to AVC. The Agatston method was used to quantify AVC, and AVC scores were standardized across MESA sites using a calcium phantom.²¹ Previous studies have demonstrated robust reproducibility between multi-detector and electron-beam CT for AVC quantification, with low intrareader and interscan variabilities of 4.4% and 9.7%.²² Aortic valve calcium scoring on CT scans was performed at MESA Visit 2 (2002–04), Visit 3 (2004–05), or Visit 5 (2010–11). Among individuals with AVC = 0 at MESA Visit 1, 89% had at least one follow-up CT scan with AVC scoring between 2 and 10 years later at MESA Visits 2, 3, or 5. Incident AVC was defined as an AVC score > 0 at the earliest MESA follow-up visit (2, 3, or 5).

Adjudication of severe aortic stenosis

Follow-up telephone interviews with participants or their proxy were conducted approximately every 9–12 months to identify new hospitalizations and diagnoses. Multi-Ethnic Study of Atherosclerosis staff requested information on hospital admissions for any reason, outpatient CVD diagnoses, and death. Death certificates were obtained from state vital statistics or the National Death Index. Multi-Ethnic Study of Atherosclerosis staff sought International Classification of Disease (ICD) diagnosis codes for all hospitalizations and requested medical records for those with ICD codes related to CVD. Multi-Ethnic Study of Atherosclerosis staff also sought to obtain outpatient records for potential CVD events. The MESA coordinating centre identified ICD codes related to aortic valve disease.

Participant follow-up data were available through 2017. Severe AS was identified using ICD coding by two cardiologists from the MESA CVD events adjudication committee independently reviewing candidate cases. Severe AS was defined according to standard criteria from (i) echocardiography (e.g. peak velocity ≥ 4.0 m/s, mean gradient ≥ 40 mmHg, aortic valve area < 1.0 cm², and so on); (ii) aortic valve replacement (surgical or transcatheter) for documented severe AS; (iii) aortic valve replacement for moderate AS when part of coronary artery bypass graft surgery; or (iv) clinical documentation of severe AS.

Aspirin intake, lipoprotein(a), and standard lipid panel

Aspirin use and frequency were measured via a questionnaire. Similar to prior studies,^{11,12} regular aspirin intake was defined as aspirin use of at least 3 days per week. A latex-enhanced turbidimetric immunoassay (Denka Seiken) was used to measure Lp(a) mass concentration in mg/dL.²³ Thresholds of Lp(a) < 50 , ≥ 50 , ≥ 75 , and ≥ 100 mg/dL were used to define increasing severity of elevated Lp(a). Total cholesterol and HDL cholesterol (HDL-C) were quantified via enzymatic methods using fasting blood samples.²⁴ The glycerol-banked enzymatic method was used to measure fasting plasma triglycerides.²⁵ The Friedewald equation was used to quantify calculated LDL-C.²⁶ Elevated LDL-C was defined according to thresholds of < 100 , ≥ 100 , ≥ 130 , and ≥ 160 mg/dL.

Covariables

Demographic and clinical information, including sex, race/ethnicity, education status, income, smoking status, and medication use history, were assessed using standardized survey methods.¹⁷ Anthropometric information, including height and weight, were measured with participants wearing minimal clothing and no shoes. Body mass index was calculated as each participant's weight in kilograms divided by their height in metres squared (m²). Cigarette smoking pack-years were measured for smoking status. Triplicate measures of blood pressure were measured from the brachial artery while

participants were in a seated resting position. The average of the second and third readings was recorded for systolic and diastolic blood pressure. Fasting blood glucose was quantified using a hexokinase/glucose-6-phosphate dehydrogenase method.²⁴ Type 2 diabetes was defined as a fasting blood glucose concentration of ≥ 126 mg/dL or the use of glucose-lowering medications. The kinetic Jaffe method was used to quantify serum creatinine, which was used to calculate estimated glomerular filtration rate (eGFR) using the Chronic Kidney Disease Epidemiology Collaboration equation.²⁷

Statistical methods

Study sample characteristics were presented in the overall sample as well as stratified by regular aspirin intake. Continuous variables were presented as mean $\pm$ standard deviation (SD) for normally distributed variables and median interquartile range for non-normally distributed variables. Categorical variables were presented as percentages. Differences between normally distributed continuous variables were assessed via the Student's *t*-test, and differences between non-normally distributed variables were assessed via Wilcoxon signed-rank test. The χ^2 tests were used to assess differences between categorical variables.

Incident AVC and severe AS cases were quantified and presented as both crude values (%) in the overall sample and according to regular aspirin intake at stratified Lp(a) and LDL-C categories. Incident AVC cases were also quantified according to follow-up time (per 1000-year follow-up).

Multivariable Cox proportional hazards regression was used to calculate hazard ratios (HRs) for incident AVC and severe AS in the overall sample and according to aspirin intake at prespecified Lp(a) and LDL-C thresholds. The proportional hazards assumption was satisfied and was tested by assessing the significance of time-dependent exposure and covariables concurrently. The association of regular aspirin intake with incident AVC and severe AS was assessed in the overall sample and according to Lp(a) and LDL-C strata, adjusting for age, sex, race/ethnicity, education, healthcare insurance, MESA site, pack-years of cigarette smoking, body mass index, systolic blood pressure, diastolic blood pressure, antihypertensive medication, Type 2 diabetes, HDL-C, LDL-C, triglycerides, Lp(a), statin utilization, and eGFR. We performed sensitivity analyses using the above models, (i) excluding individuals on baseline statin therapy; (ii) defining regular aspirin use as those who maintained ≥ 5 days/week aspirin across Visits 1, 2, and 3; (iii) assessing non-aspirin nonsteroidal anti-inflammatory drugs (NSAIDs) as an exposure; and (d) additionally using an age-, sex-, and race/ethnicity-matched propensity score for regular aspirin use. In *post hoc* analyses, the association between regular aspirin use and AVC progression across Visits 2, 3, and 5 was assessed among those with prevalent AVC at Visit 1.

Results

The mean age was 62.1 years, 53% were women, 39% were White, 12% were Chinese, 28% were Black, and 22% were Hispanic. At baseline, the mean LDL-C was 117 mg/dL, the median Lp(a) was 18 mg/dL [interquartile range (IQR) 8–41], 15% were on statins, and 23% reported regular aspirin intake. Individuals with regular aspirin intake generally had a higher burden of traditional CVD risk factors (Table 1). There was no significant difference in median Lp(a) values between aspirin non-users and regular aspirin users (18 vs 16 mg/dL, $P = .15$), while LDL-C values were lower for participants prescribed aspirin (112 mg/dL vs 118 mg/dL, $P < .001$).

Over a median follow-up of 8.9 years, there were a total of 412 participants who developed incident AVC among those with

AVC = 0 at baseline (8%). Among individuals who developed incident AVC, the median AVC score was 16 AU, with 86% who had AVC 1–99 and 14% who had AVC ≥ 100 . In the overall sample, individuals reporting regular aspirin intake had a crude incidence of AVC of 18.7 per 1000 person-years, and aspirin non-users had a crude incidence of AVC of 11.5 per 1000 person-years (Figure 1A). Among participants with Lp(a) ≥ 100 mg/dL, those who were prescribed aspirin had a lower incidence of AVC (Figure 1B). Individuals reporting regular aspirin intake had a higher incidence of AVC across all LDL-C categories (Figure 1C).

In multivariable models, Lp(a) values of 50–74, 75–99, and ≥ 100 mg/dL were associated with a 1.4–1.8-fold higher risk of incident AVC compared with Lp(a) < 50 mg/dL (P -trend = .002). In contrast, there was no significant trend towards higher risk of incident AVC across increasing LDL-C values (P -trend = .26).

After adjusting for traditional risk factors, no significant association between regular aspirin use and incident AVC was observed in the overall sample of participants (HR 1.06, 95% CI .84–1.33). However, among participants prescribed aspirin, there was a significantly lower risk of incident AVC for those with Lp(a) ≥ 75 mg/dL (HR .42, 95% CI .19–.93) and those with Lp(a) ≥ 100 mg/dL (HR .17, 95% CI .04–.67). There was no significant association between regular aspirin intake and incident AVC among persons with Lp(a) ≤ 50 mg/dL or at a threshold of Lp(a) ≥ 50 mg/dL. No significant association between regular aspirin use and incident AVC was observed across any of the LDL-C categories (Table 2).

Over a median 16.7-year follow-up, there were 86 individuals (1.3%) who developed severe AS. Regular aspirin use was associated with an 87% lower risk of incident severe AS for individuals with Lp(a) ≥ 50 mg/dL (HR .13, 95% CI .04–.47) and an even lower hazard of .02 (95% CI .001–.29) for participants with Lp(a) ≥ 75 mg/dL, whereas no significant association was observed among persons with Lp(a) < 50 mg/dL (HR 1.24, 95% CI .72–2.14). No significant association between aspirin use and severe incident AS was observed across LDL-C values (Table 3).

After excluding individuals reporting statin therapy at baseline, there were similar results by both Lp(a) and LDL-C categories for incident AVC and severe AS (see Supplementary data online, Tables S1 and S2). Results remained consistent after defining regular aspirin use as ≥ 5 days per week and across MESA Visits 1, 2, and 3 (see Supplementary data online, Tables S3 and S4). There were no significant associations of non-aspirin NSAID use with incident AVC or AS (see Supplementary data online, Tables S5 and S6). The protective association of regular aspirin use for incident AVC and AS for individuals with elevated Lp(a) remained consistent when repeating all models using a Firth bias correction (see Supplementary data online, Tables S7 and S8), additionally adjusting for age, sex, and race/ethnicity propensity scores for regular aspirin use (see Supplementary data online, Table S9) and reducing covariables (see Supplementary data online, Tables S10 and S11). There was no significant association between regular aspirin use and AVC progression (see Supplementary data online, Table S12).

Discussion

This is the first study to our knowledge that assesses the role of aspirin in CAVD according to Lp(a) and LDL-C values. We

Table 1 Characteristics of Multi-Ethnic Study of Atherosclerosis participants with measurements of aspirin, lipoprotein(a), and LDL cholesterol at Visit 1

Variable ^a	All (n = 6598)	Without regular aspirin Use (n = 5090)	Regular aspirin use (n = 1508)	P-value
Age, years	62.1 ± 10.2	61.0 ± 10.2	66.1 ± 9.2	<.001
Women, %	52.9	55.2	45.4	<.001
Ethnicity, %				<.001
White	38.5	34.9	50.7	
Chinese	11.8	13.0	8.0	
Black	27.8	28.3	25.7	
Hispanic	21.9	23.8	15.5	
Post-high school education, %	82.0	81.1	85.1	<.001
Healthcare insurance, %	91.1	89.7	95.8	<.001
Type 2 diabetes, %	12.3	10.9	17.2	<.001
Cigarette smoking, pack-years, median (Q1, Q3)	0 (0, 15)	0 (0, 14)	1 (0, 20)	<.001
Body mass index, kg/m ²	28.3 ± 5.5	28.3 ± 5.6	28.5 ± 5.2	.15
Systolic blood pressure, mmHg	126.4 ± 21.5	125.5 ± 21.4	129.5 ± 21.5	<.001
Diastolic blood pressure, mmHg	71.9 ± 10.2	71.9 ± 10.3	71.9 ± 9.9	.93
Antihypertensive medication, %	33.2	28.8	47.9	<.001
HDL cholesterol, mg/dL	51.1 ± 14.7	51.2 ± 14.7	50.8 ± 14.8	.34
Triglycerides, mg/dL, median (Q1, Q3)	110.0 (77.0, 158.0)	110.0 (77.0, 158.0)	110.0 (78.0, 158.0)	.89
LDL cholesterol, mg/dL	117.1 ± 31.4	118.4 ± 31.5	112.7 ± 30.6	<.001
Lipoprotein(a), mg/dL, median (Q1, Q3)	17.5 (7.6, 40.8)	17.8 (7.7, 40.9)	16.3 (7.1, 40.6)	.15
Statin utilization, %	14.7	11.5	25.3	<.001
Estimated glomerular filtration rate, mL/min/1.73m ²	77.8 ± 16.2	79.1 ± 16.2	73.4 ± 15.6	<.001
Prevalent aortic valve calcium, %	13.5	11.6	20.0	<.001
Incident aortic valve calcium, % ^b	8.1	9.4	11.6	<.001
Incident severe aortic stenosis, %	1.3	1.1	2.1	<.001

^aAll continuous variables presented as mean ± standard deviation unless otherwise noted.

^bAmong those with baseline aortic valve calcium = 0 (n = 5066).

observed that participants who reported regular aspirin use and had an elevated Lp(a) had a ≥60% lower risk of incident AVC and severe AS compared with persons who did not report regular aspirin use with a stepwise lower HR for those with Lp(a) ≥ 75 and ≥100 mg/dL. Contrastingly, there was no significant association between aspirin use and incident AVC across LDL-C categories. Accordingly, the results of this exploratory analysis of observational prospective cohort data suggest that aspirin may be beneficial for the prevention of CAVD for persons with elevated Lp(a) (*Structured Graphical Abstract*).

Currently, there are no approved medications to prevent AS, as prior trials²⁸ testing LDL-C lowering therapies have had null results, leaving CAVD as one of the only CVD without a proven preventive therapy. Our results are timely, as there is great enthusiasm for novel Lp(a)-lowering therapies as a preventive treatment for CAVD and the Lp(a)FRONTIERS-CAVS trial (NCT05646381) is currently beginning recruitment to test the

efficacy of pelacarsen to slow the progression of AS. However, understanding whether treatment with aspirin targeting the platelet pathway of AVC may be beneficial is of particular importance, given the wide availability and low cost of aspirin. Additionally, newer Lp(a) therapies are anticipated to be of very high cost, and there are no Food and Drug Administration therapies currently approved for AS. While treatment of anti-fibrinolytic pathways mediated by Lp(a) have not yet been explored in AS trials, our group has shown that individuals with elevated Lp(a) reporting regular aspirin intake have a lower risk of incident CVD¹¹ and CVD mortality.¹⁴ This study joins prior studies^{11–16} suggesting that aspirin use for persons with high Lp(a) may be beneficial. Further strengthening the aspirin-Lp(a)-pro-thrombotic pathophysiology link, we did not observe a significant association of regular aspirin use with incident AVC or severe AS across LDL-C, suggesting that the potential protective role of aspirin in preventing CAVD

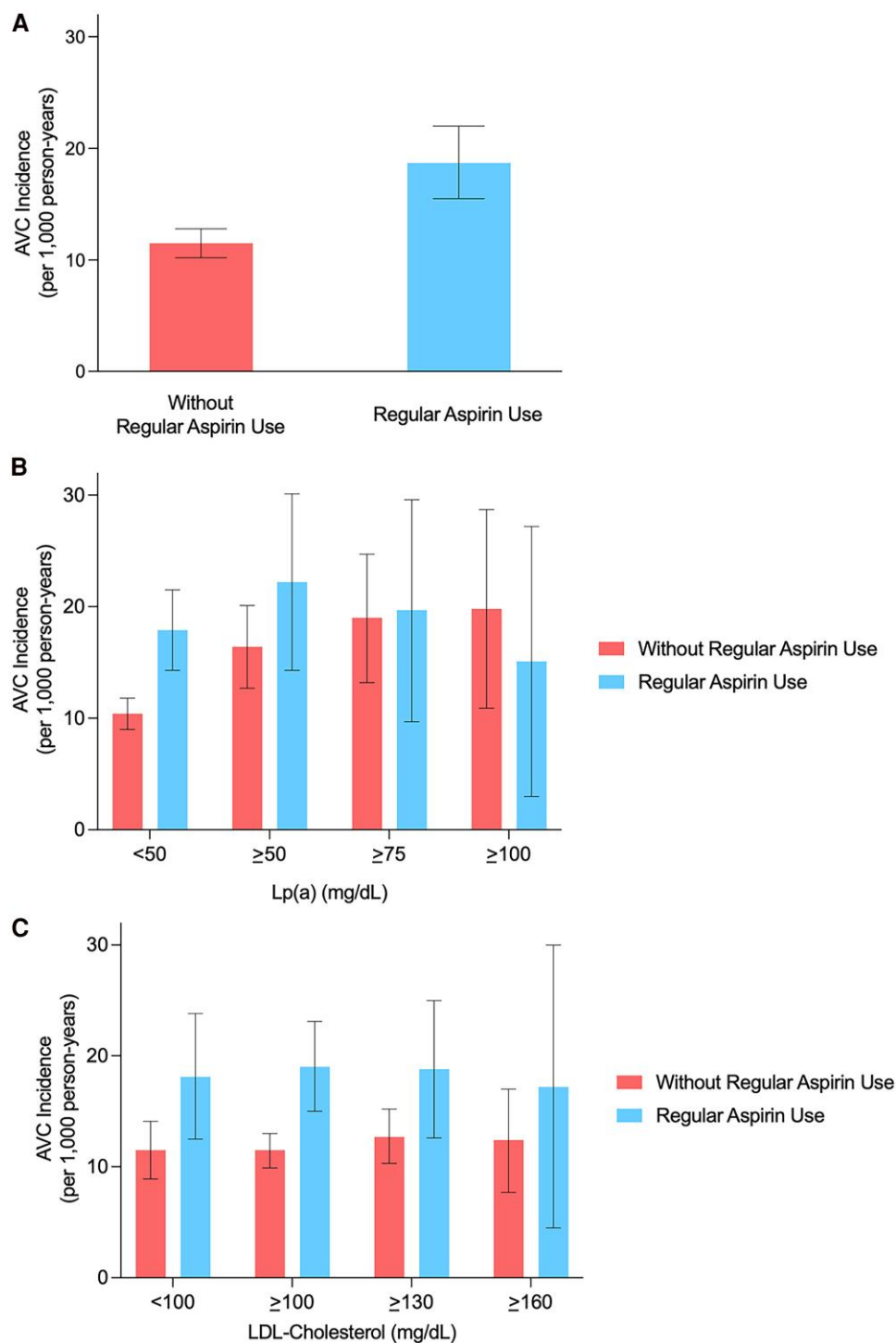


Figure 1 Crude incidence of aortic valve calcium for aspirin vs no aspirin in the overall sample (A), lipoprotein(a) (B), and LDL cholesterol (C) thresholds. *Among participants with lipoprotein(a) ≥ 100 mg/dL, those who were prescribed aspirin had a lower incidence of aortic valve calcium

may be intrinsic to the pro-thrombotic properties of Lp(a). Our findings contrast with a prior study among participants in MESA and the Heinz Nixdorf Recall Study that did not show benefit for aspirin and AVC. However, results in that prior study were not stratified by Lp(a) and had limited follow-up time.²⁹

We are the first to report on the association of aspirin with AVC and severe AS across Lp(a) values. Our HR estimates for the protective association of aspirin with AVC and severe AS may help to facilitate sample size calculations for possible future randomized controlled trials among individuals with Lp(a).

Table 2 Multivariable association of regular aspirin use with incident aortic valve calcium, across lipoprotein(a) and LDL cholesterol values

	Incident aortic valve calcium, n	HR (95% confidence interval) ^a	P-value
All (n = 5066)			
Regular aspirin use (n = 1080)	412	1.06 (.84, 1.33)	.62
Lipoprotein(a) strata			
<50 mg/dL (n = 4126)	308		
Regular aspirin use (n = 863)		1.11 (.86, 1.44)	.42
≥50 mg/dL (n = 940)	104		
Regular aspirin use (n = 217)		.99 (.61, 1.62)	.97
≥75 mg/dL (n = 477)	57		
Regular aspirin use (n = 115)		.42 (.19, .93)	.03
≥100 mg/dL (n = 214)	25		
Regular aspirin use (n = 59)		.17 (.04, .67)	.01
LDL cholesterol strata			
<100 mg/dL (n = 1456)	116		
Regular aspirin use (n = 373)		1.07 (.70, 1.62)	.76
≥100 mg/dL (n = 3610)	296		
Regular aspirin use (n = 707)		1.10 (.84, 1.45)	.50
≥130 mg/dL (n = 1628)	141		
Regular aspirin use (n = 297)		1.02 (.66, 1.58)	.92
≥160 mg/dL (n = 438)	34		
Regular aspirin use (n = 59)		1.51 (.53, 4.28)	.44

^aModels include age, sex, race/ethnicity, education, healthcare insurance, Multi-Ethnic Study of Atherosclerosis site, pack-years of cigarette smoking, body mass index, systolic blood pressure, antihypertensive medication, Type 2 diabetes, HDL cholesterol, LDL cholesterol, triglycerides, lipoprotein(a), statin utilization, and estimated glomerular filtration rate.

Additional observational studies that include a larger number of participants and assess different antiplatelet agents will also be beneficial in this regard. Beyond future studies studying antiplatelet therapies for CAVD, our findings provide added support for current initiatives for universal Lp(a) testing.³⁰

There are several mechanisms by which Lp(a) may both positively and negatively regulate platelet function.⁹ Aspirin may prevent formation of Lp(a)-mediated microthrombi on aortic valve leaflets that have a propensity to calcify by inhibiting p38 mitogen-activated protein kinase (p38MAPK), a protein linked to several functions including activation of platelet aggregation.^{31,32} Prior *in vitro* mechanistic research has suggested that calcification of interstitial aortic valve cells is induced by select proteins, including p38MAPK, and that inhibition of p38MAPK reduces Lp(a)-mediated aortic valve interstitial cell calcification.³³ However, the biological pathways implicated in the potential role of aspirin in preventing Lp(a)-mediated CAVD are complex, and further study of the underlying pathophysiology is needed.

There were several strengths of the current study. First, we included outcomes across the spectrum of CAVD, from the initial development of AVC to severe AS. There was a long period of follow-up for incident AVC (repeat CT scans were performed

over a period of 9 years) and severe AS (median 17-year follow-up). Second, our study included measures of both Lp(a) and LDL-C, in order to examine the impact of the pro-thrombotic nature of Lp(a). Third, Lp(a) and AVC values were not reported to participants or their clinicians; therefore, aspirin use was independent of knowledge of Lp(a) and AVC. Lastly, we performed comprehensive multivariable analyses for severe AS and among individuals who have undergone sequential AVC scoring on cardiac CT across a range of Lp(a) and LDL-C values, after adjusting for traditional risk factors and baseline statin therapy.

An important limitation of this study is that it is observational and therefore individuals were not randomized to aspirin therapy. Accordingly, causal inferences regarding aspirin therapy, Lp(a), and CAVD cannot be established from this study. Additionally, there may be residual confounding, and LDL-C values may have been influenced by statin therapy throughout the follow-up period. Nevertheless, we observed consistent results of regular reported aspirin use with incident AVC and severe AS across Lp(a) and LDL-C values after excluding participants reporting baseline statin therapy. Furthermore, aspirin use was self-reported in the current study, and we were unable to assess different doses of aspirin. As such, the role of aspirin dosing and

Table 3 Multivariable association of regular aspirin use with severe aortic stenosis, across lipoprotein(a) and LDL cholesterol values

	Severe aortic stenosis, n	Hazard ratio (95% confidence interval) ^a	P-value
All (n = 6598)			
Regular aspirin use (n = 1508)	86	.94 (.59, 1.52)	.81
Lipoprotein(a) strata			
<50 mg/dL (n = 5268)	65		
Regular aspirin use (n = 1183)		1.24 (.72, 2.14)	.44
≥50 mg/dL (n = 1330)	21		
Regular aspirin use (n = 325)		.13 (.04, .47)	.002
≥75 mg/dL (n = 698)	14		
Regular aspirin use (n = 182)		.02 (.001, .29)	.006
LDL cholesterol strata			
<100 mg/dL (n = 1913)	21		
Regular aspirin use (n = 510)		2.01 (.77, 5.26)	.15
≥100 mg/dL (n = 4685)	65		
Regular aspirin use (n = 998)		.70 (.39, 1.26)	.23
≥130 mg/dL (n = 2130)	22		
Regular aspirin use (n = 418)		.46 (.14, 1.47)	.19

^aModels include age, sex, race/ethnicity, education, healthcare insurance, Multi-Ethnic Study of Atherosclerosis site, pack-years of cigarette smoking, body mass index, systolic blood pressure, antihypertensive medication, Type 2 diabetes, HDL cholesterol, LDL cholesterol, triglycerides, lipoprotein(a), statin utilization, and estimated glomerular filtration rate.

its effect on Lp(a), CAVD risk, and major bleeding requires further study. However, in the ADAPTABLE (Aspirin Dosing: A Patient-Centric Trial Assessing Benefits and Long-Term Effectiveness) trial, there was no significant difference in atherosclerotic CVD or major bleeding events between individuals randomized to 81 mg vs 325 mg per day.³⁴ Additionally, 85% of participants in the ADAPTABLE trial were taking the 81 mg dose of aspirin prior to trial enrolment.³⁴ Not all MESA participants had uniform follow-up time for repeat AVC scoring, and the total number of participants with data on AVC progression was relatively low, which also limited our ability to study AVC progression. In addition, this study focused on the association of aspirin with CAVD, but if a future trial is conducted, it would be important to also consider major bleeding events and the potential net benefit of aspirin. Assessment of major bleeding events in this setting may be especially important because of the older age of CAVD onset, and older adults are a patient population at increased bleeding risk.

Conclusion

In this exploratory observational analysis, among individuals with elevated Lp(a), those who reported regular aspirin had a ≥60% lower long-term risk of incident AVC and severe AS. However, no protective association of regular aspirin intake with AVC or severe AS was observed across LDL-C values. Further studies are necessary to further elucidate the potential

beneficial role of low-dose aspirin therapy for the prevention of CAVD among persons with high Lp(a).

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Supplementary data

Supplementary data are available at [European Heart Journal](#) online.

Declarations

Disclosure of Interest

H.S.B. is a consultant/advisor for Kaneka, Novartis, Arrowhead, and Abbott. S.T. is a co-inventor and receives royalties from patents owned by UCSD and is a co-founder and has an equity interest in Oxitope and Kleanthi Diagnostics, is a consult to Novartis, and has a dual appointment at UCSD and Ionis Pharmaceuticals. The terms of this arrangement have been reviewed and approved by the University of California, San Diego, in accordance with its conflict of interest policies. M.J.B. is a member of an advisory board for Novartis, New Amsterdam, Eli Lilly, Novo Nordisk, Bayer, AstraZeneca, Boehringer Ingelheim, Genentech, Idorsia,

Agepha, and Vectura. The other co-authors have nothing to disclose.

Data Availability

All MESA data can be accessed on <https://biolincc.nhlbi.nih.gov/home/>.

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Ethical Approval

All study participants provided written informed consent at each study visit, and study protocols were approved at each MESA participating institution's local IRB and sponsored by the National Heart, Lung, and Blood Institute (<http://www.mesa-nhlbi.org>).

Pre-registered Clinical Trial Number

Not applicable.

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